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MTH 422A

1) $x_1, \ldots, x_n \mid \mu, \sigma^2 \overset{iid}{\sim} N(\mu, \sigma^2)$

$\sigma^2 \sim IG(a, b)$

$\mu \mid \sigma^2 \sim N(0, c\sigma^2)$

$f(\mu, \sigma^2 \mid \underline{x})$

$\propto (\sigma^2)^{-n/2} \exp\left(-\frac{\Sigma(x_i-\mu)^2}{2\sigma^2}\right)$

$\times (\sigma^2)^{-a-1} \exp\left(\frac{-b}{\sigma^2}\right) \times (c\sigma^2)^{-1/2} \exp\left(\frac{-\mu^2}{2c\sigma^2}\right)$

$= (\sigma^2)^{-a-\frac{n}{2}-\frac{3}{2}} \exp\left(\frac{-1}{\sigma^2}\left(\frac{\Sigma(x_i-\mu)^2}{2} + b + \frac{\mu^2}{2c}\right)\right)$

$\sigma^2 \mid \mu, \underline{x} \sim IG\Big(\underbrace{a + \frac{n}{2} + \frac{1}{2}}_{\to A}, \underbrace{\frac{\Sigma(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b}_{\to B}\Big)$

$\mu \mid \underline{x} \propto \int_0^\infty (\sigma^2)^{-A-1} \exp\left(-\frac{B}{\sigma^2}\right) d\sigma^2$

$\propto \int_0^\infty \frac{B}{t^4}\left(\frac{B}{t^2}\right)^{-A-1} \exp(-t^2)\, dt$

$= B^{-A} \int_0^\infty \left(\frac{1}{t^2}\right)^{-A+1} \exp(-t^2)\, dt$

$= B^{-A} \int_0^\infty u^{-1/2}\, u^{A-1} \exp(-u)\, du$

$= B^{-A} \int_0^\infty u^{A-\frac{1}{2}-1} \exp(-u)\, du$

let $\frac{B}{\sigma^2} = t^2$

$-\frac{B}{\sigma^4} d\sigma^2 = 2t\, dt$

$\to -\frac{t^4}{B} d\sigma^2 = 2t\, dt$

$\to d\sigma^2 = \frac{-2B}{t^4}\, dt$

let $t^2 = u$

$\to 2t\, dt = du$

$\to dt = (u)^{-1/2}\, du$

$$= \beta^{-A} \Gamma\left(A - \frac{1}{2}\right)$$

$$\propto \left(\frac{\Sigma(x_i - \mu)^2}{2} + \frac{\mu^2}{2c} + b \right)^{-\cancel{A} - \frac{1}{2} - \frac{n}{2}}$$

$= \cancel{\downarrow}$ Similar to student-t dist. with degrees of freedom $2\cancel{A} + \cancel{M} + 1$

$2a + n + 1$

$Y \sim A \nabla$

$$PPD \propto \int_{-\infty}^{\infty} \int_{0}^{\infty} (\sigma^2)^{-a - \frac{n}{2} - \frac{3}{2}} \exp\left(-\frac{1}{\sigma^2}\left(\frac{\Sigma(x_i - \mu)^2}{2} + \frac{\mu^2}{2c} + b\right)\right) \times (\sigma^2)^{-1/2} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) d\sigma^2 \, d\mu$$

$$= \int_{-\infty}^{\infty} \int_{0}^{\infty} (\sigma^2)^{-a - \frac{n}{2} - 2} \exp\left(-\frac{1}{\sigma^2}\left(\frac{\Sigma(x_i - \mu)^2}{\cancel{2}} + \frac{\mu^2}{2c} + b\right)\right) d\sigma^2 \, d\mu$$

$$= \int_{-\infty}^{\infty} \left(\frac{\sum_{i=1}^{n+1} (x_i - \mu)^2}{2} + \frac{\mu^2}{2c} + \cancel{b} \right)^{-a - \frac{n}{2} - 2} d\mu$$

assuming large c & small b & a

$$= \cancel{\int_{-\infty}^{\infty} \left(\frac{\sum_{i=1}^{n+1} (x_i - \mu)^2}{2} \right)^{-a - \frac{n}{2} - 2} d\mu}$$

$$= \cancel{\int_{-\infty}^{\infty} t^2}$$

let $\cancel{\Sigma (\mu - x_i)^2 = t}$

$$\frac{\sum_{i=1}^{n+1} (x_i - \mu)^2}{2} = t^2$$

$$= \int_{-\infty}^{\infty} \left(\frac{\Sigma(x_i - \mu)^2}{2b} + \frac{\mu^2}{2cb} + 1 \right)^{-a - \frac{n}{2} - 2} d\mu$$

Date
Page

$$= \int_{-\infty}^{\infty} \left(\frac{\sum (x_i - \mu)^2}{n} \right)^{-\frac{n}{2} - 2} d\mu$$

$$= \int_{-\infty}^{\infty} \left(\sum x_i^2 - \mu \sum x_i + \mu^2 + (\sum x_i)^2 - (\sum x_i)^2 \right)$$

$$= \int_{-\infty}^{\infty} \left[\sum x_i^2 - (\sum x_i)^2 \right]^{-\frac{n}{2}-2} \left(1 + \frac{(\mu - \sum x_i)^2}{\sum x_i^2 - (\sum x_i)^2} \right)^{-2} d\mu$$

$$= \int_{-\infty}^{\infty} \left[\sum x_i^2 - (\sum x_i)^2 \right]^{-\frac{n}{2}-2} \left(1 + \frac{\left(\frac{\mu - \sum x_i}{\sqrt{\sum x_i^2 - (\sum x_i)^2 / (n-4)}} \right)^2}{n-4} \right)^{-\frac{n}{2}-2} d\mu$$

$$= \left[\sum x_i^2 - (\sum x_i)^2 \right]^{-\frac{n+4}{2}} \int_{-\infty}^{\infty} \left(1 + \frac{\left(\frac{(\mu - \sum x_i)}{\sqrt{\sum x_i^2 - (\sum x_i)^2 / (n-4)}} \right)^2}{n-4} \right)^{-\frac{n-4}{2}} d\mu$$

let $\dfrac{\mu - \sum x_i}{\sqrt{\sum x_i^2 - (\sum x_i)^2 / n-4}} = t$

$\rightarrow \quad d\mu = \sqrt{\dfrac{\sum x_i^2 - (\sum x_i)^2}{n-4}}\, dt$

$$\propto \left[\sum x_i^2 - (\sum x_i)^2 \right]^{-\frac{n+2}{2}}$$

$$\propto \left[\cancel{x_{n+1}^2} - \left(\sum_{i=1}^{n} x_i\right) x_{n+1} - x_{n+1}^2 + \sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2 \right]^{-\frac{n+2}{2}}$$

$$= \left[\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2 - 4 \cdot x_{n+1} \sum_{i=1}^{n} x_i \right]^{-\left(\frac{n+2}{2}\right)}$$

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2) $X_1, \dots, X_m \mid \mu_1, \sigma^2 \overset{IID}{\sim} N(\mu_1, \sigma^2)$

$Y_1, \dots, Y_n \mid \mu_2, \sigma^2 \overset{IID}{\sim} N(\mu_2, \sigma^2)$

$\sigma^2 \sim IG(a, b)$

$\mu_1, \mu_2 \mid \sigma^2 \sim N(0, c\sigma^2)$

So, $\mu_1 - \mu_2 \sim N(0, 2c\sigma^2)$

$p(\mu_1 - \mu_2, \sigma^2 \mid \underline{X}, \underline{Y})$

$$\propto (\sigma^2)^{-m/2} \exp\left[-\frac{1}{2}\left(\frac{\sum_{i=1}^{m}(x_i-\mu_1)^2}{\sigma^2}\right)\right](\sigma^2)^{-n/2} \exp\left[-\frac{1}{2}\frac{\sum_{i=1}^{n}(y_i-\mu_2)^2}{\sigma^2}\right]$$

$$(\sigma^2)^{-a-1} \exp\left(-\frac{b}{\sigma^2}\right) (\sigma^2)^{-1/2} \exp\left(\frac{(\mu_1-\mu_2)^2}{4c\sigma^2}\right)$$

$$= (\sigma^2)^{-\left(\frac{m+n+2a+\cancel{1}+1}{2}\right)} \exp\left[-\frac{1}{\sigma^2}\left(\frac{\sum_{i=1}^{m}(x_i-\mu_1)^2}{2} + \frac{\sum_{i=1}^{n}(y_i-\mu_2)^2}{2}\right.\right.$$

$$\left.\left. + b + \frac{(\mu_1-\mu_2)^2}{4c}\right)\right]$$

$$= (\sigma^2)^{\underbrace{-\left(\frac{m+n+2a+\cancel{1}+1}{2}\right)}_{A-1}} \exp\left[-\frac{1}{\sigma^2}\underbrace{\left(\frac{\sum_{i=1}^{m}(x_i-\mu_1)^2}{2} + \frac{\sum_{i=1}^{n}(y_i-\mu_2)^2}{2} + \frac{(\mu_1-\mu_2)^2}{4c} + b\right)}_{B}\right]$$

$\cancel{\mu_1 - \mu_2, \sigma^2 \mid X, Y \sim IG}\left(\cancel{\frac{m+n+2a}{2}}, B\right)$ Similar to $IG(A, B)$

$$\pi(\mu_1 - \mu_2 \mid \underline{X}, \underline{Y}) = \int_0^\infty (\sigma^2)^{-\left(\frac{m+n+2a+1}{2}\right)-1} \exp\left(\frac{\sum_{i=1}^{m}(x_i-\mu_1)^2}{2} + \frac{\sum_{i=1}^{n}(y_i-\mu_2)^2}{2} + \frac{(\mu_1-\mu_2)^2}{4c}\right) d\sigma^2$$

~~exp~~

$$\propto \left(\frac{\sum_{i=1}^{m}(x_i-\mu_1)^2}{2} + \frac{\sum_{i=1}^{n}(y_i-\mu_2)^2}{2} + \frac{(\mu_1-\mu_2)^2}{4c} + b\right)^{-A}$$

$$A = \frac{m+n+2a+1}{2}$$

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3) $X_1, \ldots, X_m \overset{IID}{\sim} \text{Poisson}(\lambda_1)$

$Y_1, \ldots, Y_n \sim \text{Poisson}(\lambda_2)$

$\lambda_1, \lambda_2 \overset{IID}{\sim} \text{Gamma}(a, b)$

$$p(\lambda_1, \lambda_2 \mid \underline{X}, \underline{Y}) \propto \lambda_1^{\sum^m x_i} \lambda_2^{\sum^n y_i} \exp(-m\lambda_1) \exp(-n\lambda_2) \times \lambda_1^{a-1} \exp(-b\lambda_1) \times \lambda_2^{a-1} \exp(-b\lambda_2)$$

$$= \lambda_1^{a+\sum^m x_i - 1} \lambda_2^{a+\sum^n y_i - 1} \exp(-(m+b)\lambda_1) \exp(-(b+n)\lambda_2)$$

$\lambda_1 \mid \underline{X}, \underline{Y} \sim \text{Gamma}(a + \sum_{i=1}^{m} X_i, m+b)$

$\lambda_2 \mid \underline{X}, \underline{Y} \sim \text{Gamma}(a + \sum_{i=1}^{m} Y_i, m+b)$

Assuming $m = n$

$\lambda_1 + \lambda_2 \mid \underline{X}, \underline{Y} \sim \text{Gamma}(a + \sum^m X_i + \sum Y_i, b+n)$

$y = x_1 + x_2 \qquad \theta = \frac{x_1}{x_1+x_2}$

$x_1 = \theta y \qquad x_2 = y(1-\theta)$

$$|J| = \begin{vmatrix} \frac{\partial \theta y}{\partial y} & \frac{\partial y(1-\theta)}{\partial y} \\ \frac{\partial \theta y}{\partial \theta} & \frac{\partial y(1-\theta)}{\partial \theta} \end{vmatrix}$$

$$= |-\theta y - y + \theta y| = y$$

$$\pi(\theta, y \mid \underline{x}, \underline{y}) \propto (\theta y)^{a+\Sigma x_i - 1} (y(1-\theta))^{a+\Sigma y_i - 1}$$

$$\exp(-(b+m)\theta y) \exp(-(b+m) y(1-\theta))$$

assuming $n = m$

$$~~= (\theta^{a+\Sigma x_i + \Sigma y_i - 1} \exp(-(m+b)\theta)~~$$

$$= (y^{a+\Sigma x_i + \Sigma y_i - 1} \exp(-(m+b) y))$$
$$\times (\theta^{a+\Sigma x_i - 1} (1-\theta)^{a+\Sigma y_i - 1})$$

$$\propto \pi(y \mid \underline{x}, \underline{y}) \; \pi(\theta \mid \underline{x}, \underline{y})$$

So,

$$\theta \mid \underline{x}, \underline{y} \sim \text{Beta}\left(a + \sum_{i=1}^{m} x_i,\; a + \sum_{i=1}^{m} y_i\right)$$

$H: x_1 = x_2$ equivalent to $\theta = \frac{1}{2}$

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4) $x_1 \sim N(0,1)$

$x_{t+1} | x_t \sim N(\rho x_t, 1-\rho^2)$

$\rho \sim U(-1,1)$

$\therefore f(x_1, \dots x_T | \rho)$

$= f(x_T | x_{T-1} \dots x_1, \rho) f(x_{T-1}, x_{T-2}, \dots x_1 | \rho)$

$\vdots$

$= \left(\prod_{t=2}^{t=T} f(x_t | x_{t-1}, \dots, x_1, \rho)\right) f(x_1 | \rho)$

$= \left(\prod_{t=2}^{t=T} f(x_t | x_{t-1})\right) f(x_1 | \rho)$

$\propto (1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left(-\frac{1}{2}\left(\frac{\sum (x_t - \rho x_{t-1})^2}{1-\rho^2}\right)\right) \exp\left(-\frac{x_1^2}{2}\right)$

~~$\pi(\rho | x_1, \dots, x_T) \propto (1-\rho^2)^{-\left(\frac{T-1}{2}\right)}$~~

$p(\rho | x_1, \dots, x_T) \propto (1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left(-\frac{1}{2}\left(\frac{\sum_{t=2}^{T} (x_t - \rho x_{t-1})^2}{1-\rho^2}\right)\right)$ ~~$\exp\left(-\frac{x_1^2}{2}\right)$~~ $I(\rho \in [-1,1])$

Now, x_t

$PPD = \pi(x_{T+1}, x_{T+2} | x_1, \dots, x_T)$

~~$\propto \int (1-\rho)^{-1/2} \exp\left(-\frac{(x_{T+2} - \rho x_{T+1})^2}{2(1-\rho^2)}\right)(1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left(-\frac{1}{2}\right.$~~

$\rho \in S_\rho$

$$\propto \int_{\rho \in S_\rho} (1-\rho^2)^{-2/2} \exp\left(\frac{-(x_{t+2}-\rho x_{t+1})^2}{2(1-\rho^2)} - \frac{(x_{t+1}-\rho x_t)^2}{2}\right)(1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left(-\frac{\sum_{t=2}^{T}(x_t-\rho x_{t-1})^2}{2(1-\rho^2)}\right)$$

$$\times \exp\left(\frac{x_1^2}{2}\right) d\rho$$

$$= \int_{-1}^{1} (1-\rho^2)^{-T/2} \exp\left(-\frac{\sum_{t=2}^{T+2}(x_t-\rho x_{t-1})^2}{2(1-\rho^2)}\right) \exp\left(-\frac{x_{t+1}+x_1^2}{2}\right) d\rho$$

$$\exp\left(-\frac{x_{t+1}-\rho x_t}{2(1-\rho^2)}\right)$$

PPD =

$$PP = \int \frac{f(x_1, \ldots, x_{T+2})}{f(x_1, \ldots, x_T)}$$

$$= \int \frac{(1-\rho^2)^{-\left(\frac{T+1}{2}\right)} \exp\left(-\frac{\sum_{t=2}^{T+2} x_t - \rho x_{t-1}}{2(1-\rho^2)} - \frac{x_1^2}{2}\right)}{(1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left(-\sum_{t=2}^{T} x_t\right)}$$

$$= \int_{-1}^{1} (1-\rho^2)^{-\frac{(T+1)}{2}} \exp\left(-\frac{1}{2}\sum_{t=2}^{T+2} \frac{(x_t-\rho x_{t-1})^2}{(1-\rho^2)}\right) d\rho$$